

MACHINE LEARNING

UNIT - 01

Introduction to Machine Learning.

Machine learning is a branch of Artificial Intelligence (AI) that allows computers to learn from data and make decisions or predictions without being explicitly programmed.

Instead of writing rules, we give data, and the computers learn pattern from it.

Machine learning is a subset of Artificial Intelligence.

It is useful when

- Data is large
- Rules are complex
- Patterns are difficult to write manually.

Artificial Intelligence

Any technique that enables computers to mimic human intelligence, using logic, if-then rules, decisions trees and machine learning (including deep learning)



Machine Learning

A Subset of AI that includes that include statistical techniques that enable machines to improve at tasks with experience. This include deep learning.



Deep learning

The subset of ML that allow software to train itself to perform tasks, like speech and image recognition, by exposing multilayered neural network to large data.

Important terms of Machine learning.

- ① Algorithm → A step-by-step method used by a computer to solve a problem or learn from data.

Example → Decision Tree. Linear Regression.

- ② Dataset → A collection of data used to train and test a machine learning model.

Example → Images, numbers

- ③ Training Data → The data used to ~~check~~ teach the model so it can learn patterns.

- ④ Test Data → Data used to check how well the model works after training.

Scope of Machine Learning.

The scope means the area where machine learning can be used.

- ① Healthcare → It helps doctors detect diseases, analyze medical images, and predict patient risks.
- ② Finance and Banking → Used for fraud detection, credit scoring, and stock market prediction.
- ③ E-commerce → Recommends products to users based on their interests (like online shopping recommendations).
- ④ Self-Driving Cars → It helps vehicles to recognize roads, objects and traffic signals.
- ⑤ Natural Language Processing → Used in chatbots, language translation, and voice assistants.
- ⑥ Education → Personalized learning systems and intelligent tutoring systems.

Limitations of Machine Learning

Despite its advantages, Machine Learning also has some limitations.

- ① Needs Large Amount of Data → Machine learning models require a lot of data to learn properly.
- ② High cost → Training machine learning models requires powerful computers and infrastructure.
- ③ Time-consuming → Training and testing models can take a long time.
- ④ Data Quality Problem → If the training data is incorrect or biased, the model will give wrong results.
- ⑤ Security Risks → Machine learning systems can be attacked or manipulated with fake data.
- ⑥ Predictor Variables → It is a feature(s) of the data that can be used to predict the output.

Advantages / scope of ML.

① Automation of Tasks.

Machine learning can automatically perform tasks without human intervention, it reduces manual work and increases efficiency.

② Handles Large Data.

ML can analyze and process very large amounts of data quickly and accurately.

③ Improves with experience.

ML systems learn from data and improve their performance over time.

④ Accurate Predictions.

ML models can make accurate predictions and decisions based on patterns in data.

⑤ Wide Application.

Machine learning is used in many fields such as healthcare, finance, education, transportation, and e-commerce.

Types of Machine Learning.

→ A machine can learn to solve a problem by any one of the following three approaches:-

These are the ways in which a machine can learn:-

- ❖ Supervised learning
- ❖ Unsupervised learning
- ❖ Reinforcement learning.

1 Supervised Learning.

→ It is a type of machine learning in which the model is trained using labeled data.

→ This means the input data already has the correct output (target), and the algorithm learns the relationship between input and output.

→ The machine learns from examples with correct answers.

Example → Suppose we want to predict wheather an email is spam or not spam.

Email features
Contains many ads

Output
spam

From known person

Not spam

The machine studies many such examples and learn to classify new emails.

How supervised Learning works.

1. Training data is given with input and correct output.
2. The algorithm learns the pattern between input and output.
3. The model is trained.
4. The trained model predicts the output for new data.

Types of Supervised Learning.

1. Classification → Output is a category or class.
- Example → Spam detection, disease detection.

2. Regression $\rightarrow$ Output is a continuous value.

Example $\rightarrow$ Predicting house price or temperature.

Application of supervised learning.

- ◆ Email spam detection
- ◆ Face Recognition
- ◆ Medical Diagnosis
- ◆ Stock price predictions.
- ◆ Image classification.

2. Unsupervised Learning.

Unsupervised learning is a type of machine learning in which the model is trained using data without labeled outputs.

The algorithm analyzes the data and finds hidden patterns or structures in the dataset automatically.

Example $\rightarrow$ Suppose we have data of many customers but we do not know their categories.

The algorithm studies the data and groups of customers with similar behavior together. This process is called clustering.

Types of Unsupervised learning.

1. **Clustering**. - It groups similar data points together.

Example :- Customer segmentation

2. **Association** :- It defines and finds relationships between variables in the data.

Example → Market basket analysis (people who buy bread often buy butter).

Common Algorithm.

- ◆ K-means clustering
- ◆ Hierarchical clustering
- ◆ DBSCAN
- ◆ Apriori Algorithm.

Application.

- ① Customer segmentation
- ② Recommendation systems.
- ③ Fraud detection.
- ④ Market analysis.

Advantages.

1. No labeled data is required
2. It helps discover hidden patterns in data.
3. Useful for analyzing large datasets.

Disadvantages.

1. Results can be difficult to interpret
2. Accuracy is harder to evaluate.
3. Sometimes it may find irrelevant patterns.

③ Reinforcement Learning.

Reinforcement learning is a type of machine learning in which an agent learns by interacting with an environment. The agent performs actions and receives rewards or punishment, and it learns to choose actions that maximize the total reward over time.

Basic Component.

- 1/ Agent → A learner or decision maker.
- 2/ Environment → The place where the agent operates.

- 3] Action The choices made by the agent.
- 4] Reward Feedback given to the agent for its actions. (positive or negative)
- 5] State The current situation of the environment.

Working of Reinforcement Learning.

1. The agent observes the current state of the environment.
2. It choose an action.
3. The environment responds with a reward or penalty.
4. The agents updates its strategy to maximize future rewards.

Example.

A robot learning to walk.

- If it moves correctly, it gets a positive reward.
 - If it falls, it get negative reward.
- Over time, the robot learns the best way to walk.

Application

- Robotics
- Game playing (like chess or video games)
- Self-driving cars.
- Recommendation systems.

Advantages

- Learns from interaction with the environment.
- Can solve complex decision-making problems.
- Improves performance over time.

Disadvantages

- Requires a large amount of training time
- Designing the reward system can be difficult.
- Sometimes it may learn inefficient behaviors

Regression

REGRESSION

Regression is a method in machine learning that finds the relationship between independent variables (inputs) and a dependent variable (output) to predict a continuous value.

Regression is used when the output is a number not a category.

Example

Some common examples:-

- Predicting house price based on size and location.
- Predicting temperature for tomorrow.
- Predicting salary based on experience.
- Predicting sales of a product.

How Regression works.

1. The model takes training data (input + output)
2. It finds the mathematical relationship between them
3. It creates a best-fit line or curve
4. This line is used to predict new values

Example Formula (Linear Regression)

$$y = mx + c$$

where

y = predict output

m = slope

x = input variables

c = intercept

Types of Regression.

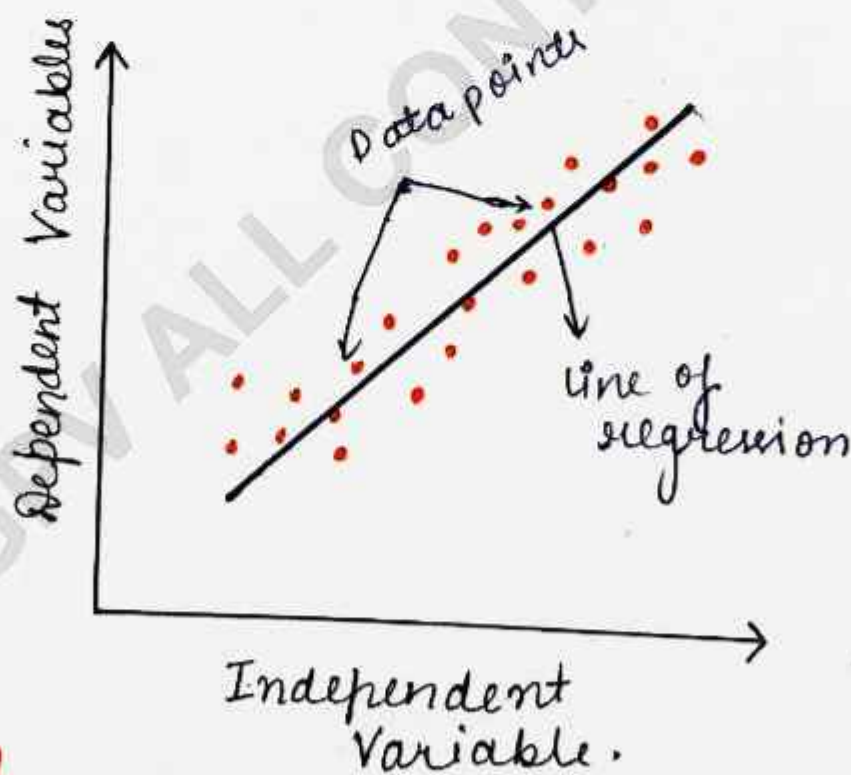
1. Linear Regression
2. Polynomial Regression
3. Support vector Regression
4. Decision tree regression
5. Random Forest regression.

1. LINEAR REGRESSION

Linear Regression is a regression algorithm used to model the linear relationship between independent variables (X) and a dependent variable Y .

It predicts values using a straight line formula.

$$Y = mX + c$$



working

1. The algorithm takes training data.
2. It finds the best-fit straight line that minimize prediction error.
3. This line is used to predict new values.

Example.

Predicting house price based on area.

Area = 1000 sq ft

Price = ₹ 30 Lakh

Linear regression finds the relationship between area and price

POLYNOMIAL REGRESSION.

Polynomial Regression is a type of regression used when the relationship between (X) and output (Y) is not linear (not a straight line) but curved.

It is actually an extension of linear regression, but it models data using polynomial equation.

Formula $\Rightarrow Y = a + bx + cx^2 + dx^3 + \dots + kx^n$

Y = predicted output

x = input variable

a, b, c, d = coefficients.

n = degree of polynomial.

Example - (degree 2)

$$Y = a + bx + cx^2.$$

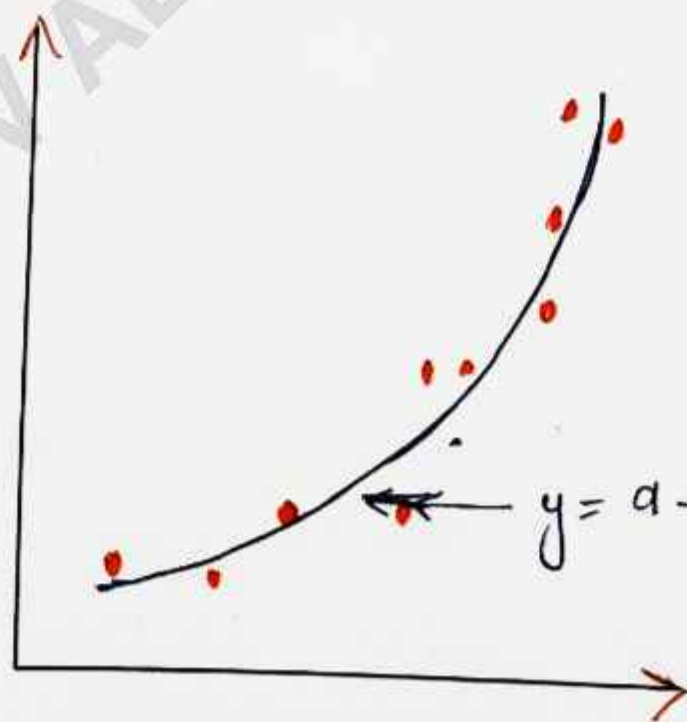
Use of Polynomial Regression.

Linear regression fails when data is curved.

Example :- Growth of plant over time
Population growth.
Temperature variation.

These follow a curve, not a straight line.

Graph →



Polynomial model.

$y = a + bx$ ~~is not a straight line~~

Working -

1. Take input data (X, Y)
2. Convert X into polynomial features
 $X \rightarrow X, X^2, X^3, \dots$
3. Apply linear regression on these new features.
4. The model fits a curve instead of a straight line
5. Use the curve to predict new values.

degree of polynomial.

The degree (n) controls the shape of the curve

Degree 1 $\rightarrow$ straight line (linear)

Degree 2 $\rightarrow$ U-shape curve (parabola)

Degree 3 $\rightarrow$ more complex curve.

SUPPORT VECTOR REGRESSION.

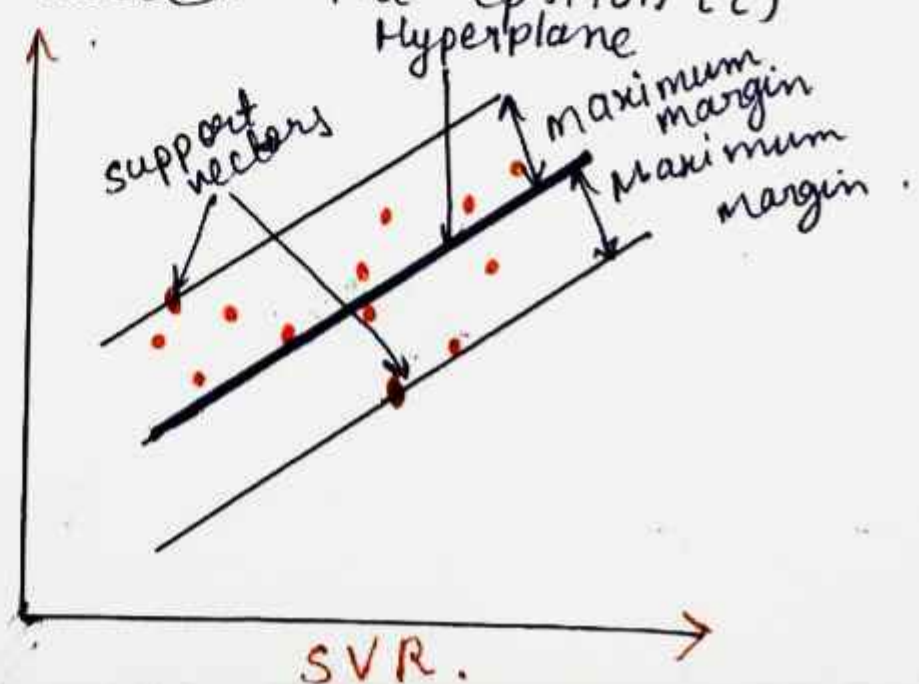
SVR is a regression algorithm based on support vector machines (SVM) that predicts continuous values by finding a function that fits the data within a fixed error margin (ϵ)

• Instead of minimizing total error, SVR tries to keep error within a limit.

SVR does not fit all data points exactly. It tries to:

- Fit a line (or curve)
- Keep most data points within a margin (ϵ)

→ This margin is called the epsilon (ϵ) tube.



DECISION TREE REGRESSION.

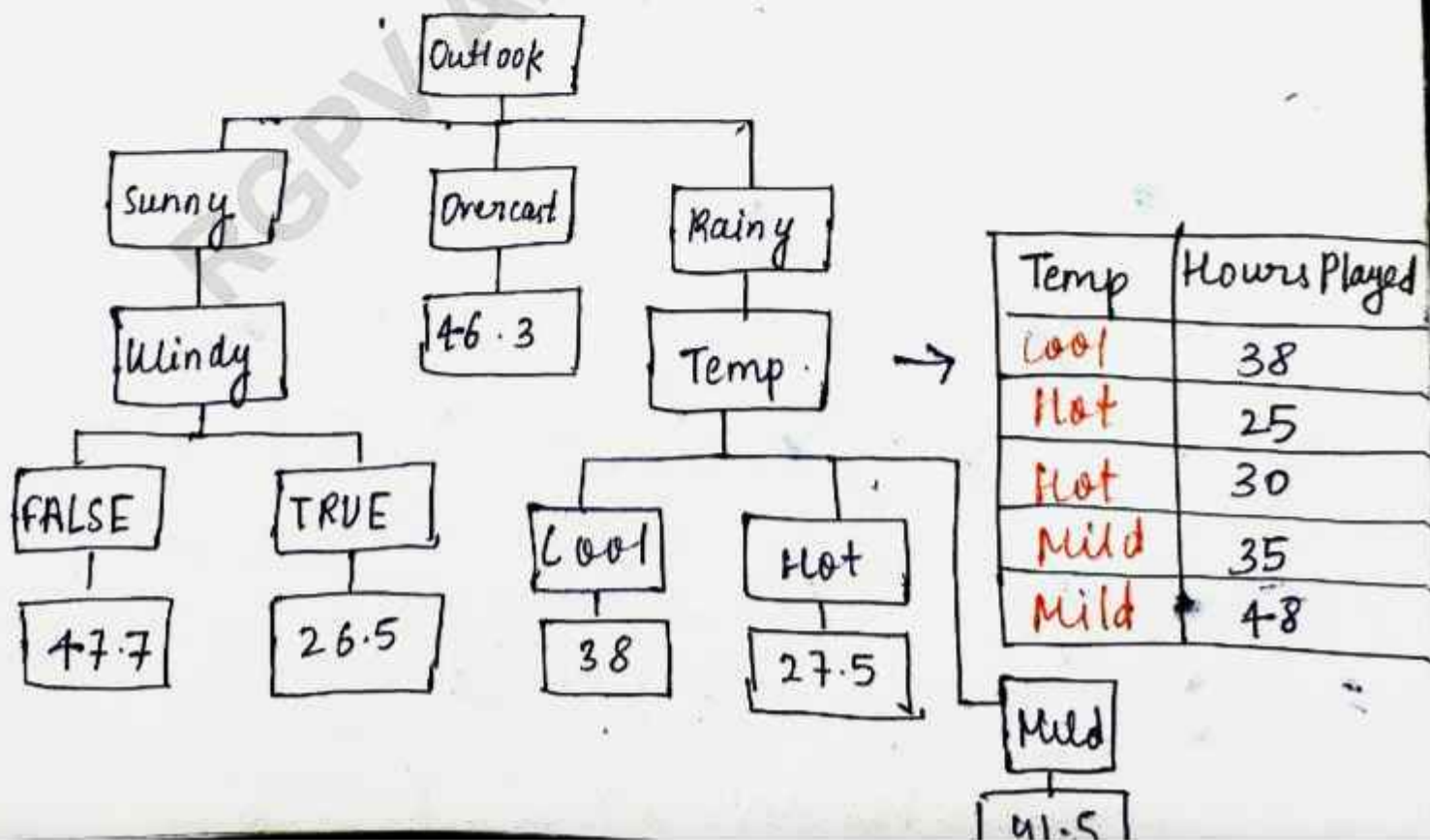
It is a machine learning algorithm that predicts continuous values by splitting the dataset into smaller parts using a tree like structure.

It creates decisions step-by-step and give a numerical output.

Instead of fitting a line (like linear regression)

- It Divides data into regions.
- Assign a constant value to each region.

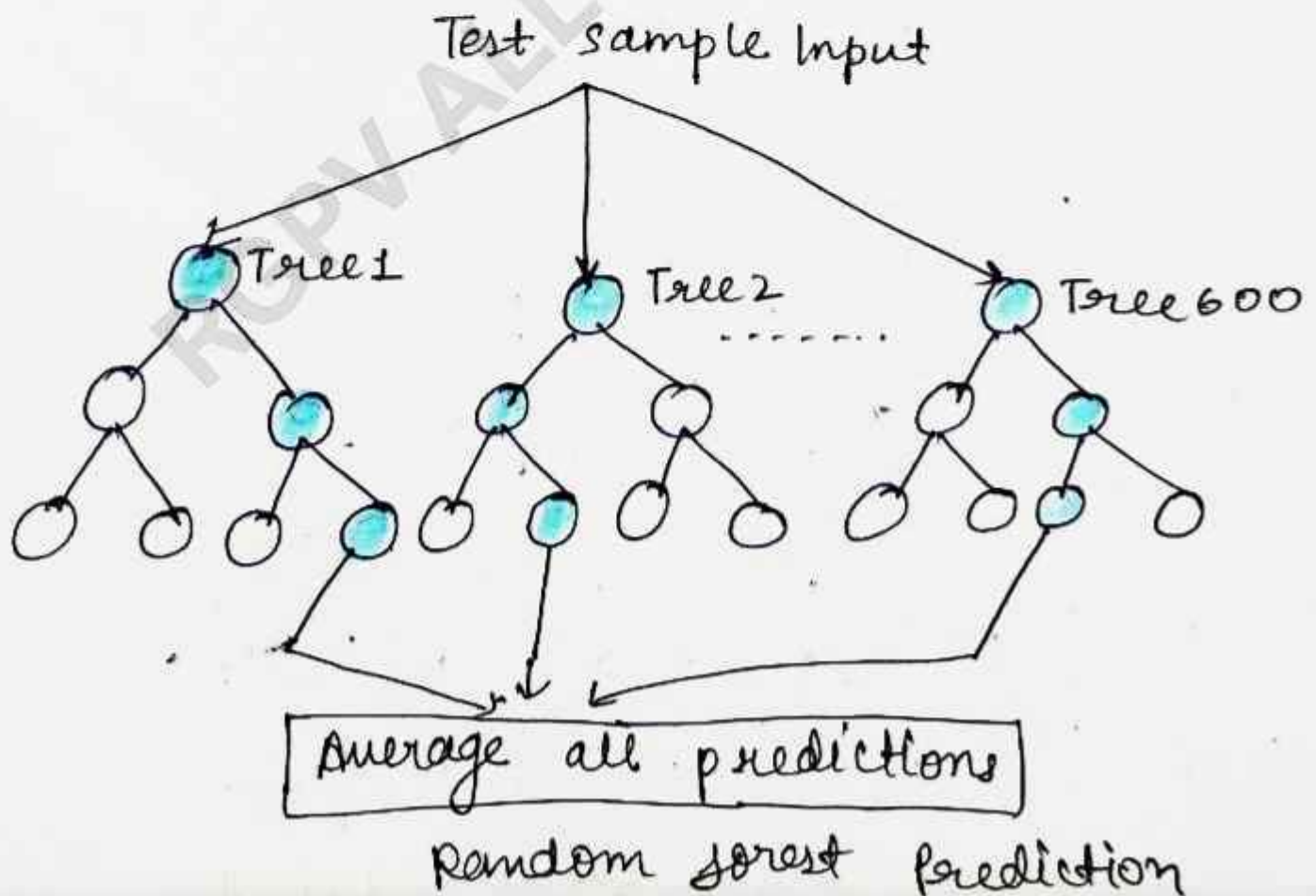
So prediction is based on which region the input falls into.



RANDOM FOREST REGRESSION

Random Forest Regression is an algorithm that predicts continuous values by combining the result of multiple decision trees.

- Instead of using one tree, it uses a forest (many trees).
- One decision tree $\Rightarrow$ may overfit.
- Many trees together $\Rightarrow$ better accuracy.
- Final prediction $\Rightarrow$ average of all tree predictions



PROBABILITY

Probability is the mathematical foundation of Machine Learning used to model uncertainty, randomness, and prediction confidence.

In real-world data: Data is noisy. Outcomes are uncertain and exact answer is not possible.

So, ML uses probability to say :
This prediction is 80% likely.

$$0 \leq P(E) \leq 1$$

0 $\rightarrow$ event will never happen

1 $\rightarrow$ event will definitely happen

Basic Probability Terminology.

① Random Experiment: An experiment whose outcome cannot be predicted exactly.

Example

- ① Tossing a coin
- ② Rolling a dice
- ③ selecting a customer randomly.

② Sample space (S) :- The set of possible outcomes of an experiment

Example $\rightarrow$ Rolling a dice.

$$S = \{1, 2, 3, 4, 5, 6\}$$

③ Event (E) - A subset of a sample space.

Example $\rightarrow$ Even (A) $\rightarrow$ getting an even number.
 $A = \{2, 4, 6\}$.

Common type of Probability.

① Classical Probability $\rightarrow$ It is used when all outcomes are equally likely.

$$P(E) = \frac{\text{Favorable outcomes}}{\text{Total outcomes}}$$

Example $\rightarrow$ Probability of getting Head in a coin toss

$$P(H) = \frac{1}{2}$$

- ② Empirical (experimental) Probability.
Based on past data and observation.

$$P(E) = \frac{\text{No. of time event occurs}}{\text{Total Trials}}$$

Example → Outcome of 100 customers, 60 purchased.

$$P(\text{Purchase}) = \frac{60}{100} = 0.6$$

- ③ Subjective Probability.
Based on personal experience or judgement.

Example → Probability of stock market rising tomorrow.

PROBABILITY DISTRIBUTION.

It describes how the values of a random variable are distributed along with their probabilities.

A distribution tells us which values occur, how often, and with what probability.

Importance →

- It helps us to
- ① Understand data behavior
 - ② Identify patterns & trends
 - ③ Make prediction and soon

STATISTICS

Statistics is the foundation of Machine Learning because it helps us understand data, make predictions, and evaluate models.

Statistics is the science of collecting, analyzing, interpreting, and presenting data.

In Machine Learning, statistics is used to

- Understand patterns in data
- Make Predictions
- Handle Uncertainty
- Evaluate Model performance.

→ In simple words:-

$ML = \text{Data} + \text{Statistics} + \text{Algorithms}.$

Types of Statistics.

- ① Descriptive data Statistics.
- ② Inferential Statistics.

① Descriptive Statistics.

Descriptive Statistics is the branch of statistics used to organize data.

Summarize data and describe the main features of data.

- It does not make predictions.
- It only tells us what the data look like.

Importance.

Before building any ML model, we must.

- Understand the dataset
- Detect patterns
- Find errors or outliers.

② Inferential Statistics:-

It is the branch of statistics that uses sample data to draw conclusions, make predictions, or take decisions about a population.

Sample → subset of population, small & manageable, denoted by n .

main component → Estimation, Hypothesis Testing, Regression analysis, Analysis of Variance.

LINEAR ALGEBRA.

Linear Algebra is a branch of mathematics that deal with vectors, matrices and linear equations.

It is very important in fields like machine learning, data science, computer graphics and Engineering.

★ Linear algebra studies :-

Vectors → quantities with direction and magnitude.

Matrices → rectangular arrays of numbers

Linear equations → equation forming straight lines.

Example of linear equation:

$$2x + 3y = 5.$$

Example of linear Algebra in Machine Learning.

- ① Dataset and Data Files.
- ② Images and Photographs.
- ③ Linear Regression.
- ④ Principle component Analysis.
- ⑤ Deep Learning.
- ⑥ Recommender system.

CONVEX OPTIMIZATION.

Convex Optimization is a very important topic in machine learning and mathematics, especially for solving problems where we want the best (optimal) solution.

Convex optimization is a type of optimization problem where:-

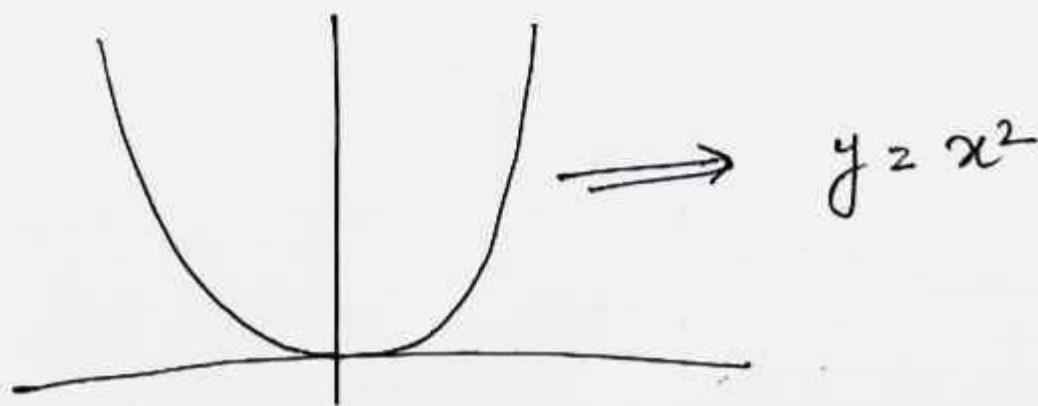
- ① The objective function is convex
- ② The feasible region is also convex.

convex function $\rightarrow$ A function is convex if the line between any two points lies above or on the curve.

Mathematical condition

$$f(\theta x_1 + (1-\theta)x_2) \leq \theta f(x_1) + (1-\theta)f(x_2)$$

Example = $f(x) = x^2$



DATA VISUALIZATION

Data Visualization is the process of transforming raw data into graphical representations like charts, graphs, and dashboards to make information.

Easy to understand, Insightful and Actionable.

It is a key part of EDA (Exploratory Data Analysis) in Machine Learning.

Need of Data Visualization.

Data visualization is used when:

① Easy understanding → Complex data becomes simple.

② Identity patterns → Trends and patterns are easy to see.

③ Better decision making → Helps in analyzing information quickly.

④ Saves times Visual data is faster to read than tables.

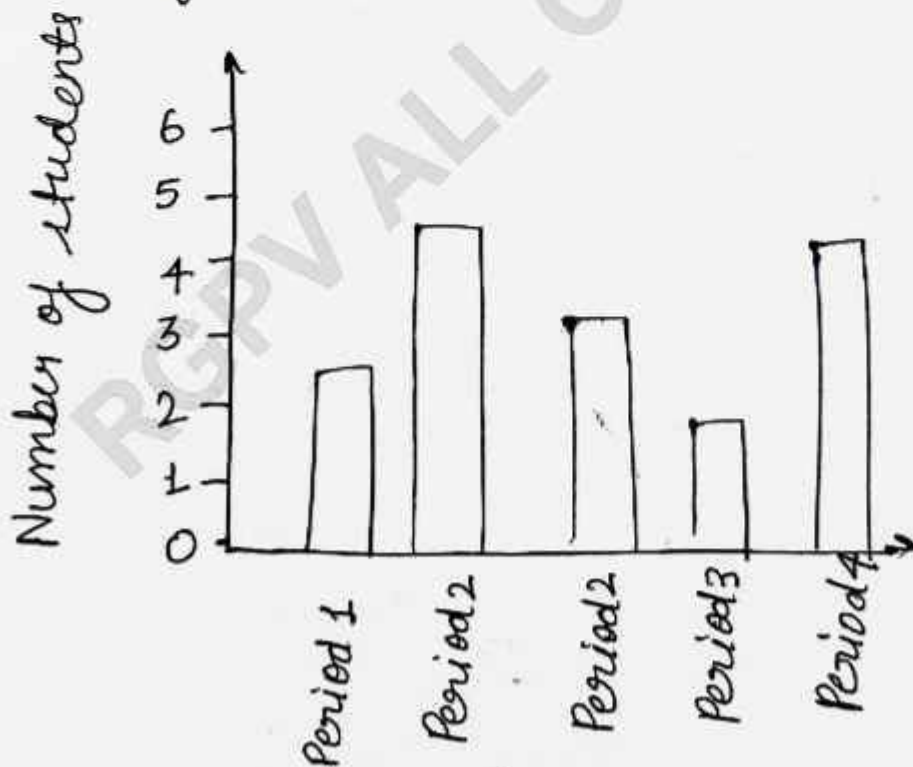
⑤ Detect errors - Outliers or unusual values become visible.

Common types of Data Visualization

- ① Bar chart
- ② Line chart
- ③ Pie chart
- ④ Histogram
- ⑤ Scatter plot.

BAR CHART

The Bar chart is a graphical representation of data using rectangular bars where the length or height of each bar represents the value of the data.



Class categories.

A bar chart has two axes :-

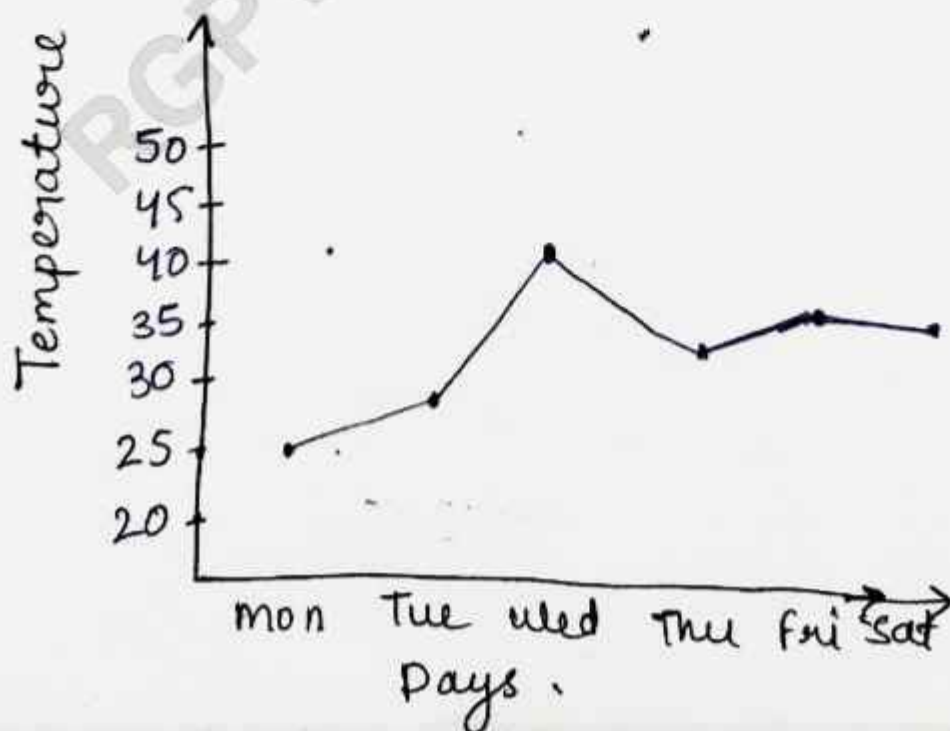
- X-axis (Horizontal axis) → shows categories (subject, products, years, etc).
- Y-axis (Vertical axis) → shows values or numbers

LINE CHART

A line chart is a type of graph that shows changes in data over time by connecting data points with a continuous line.

A line chart represents data using points connected by straight lines to show trends or changes.

Example → Temperature change during a week.

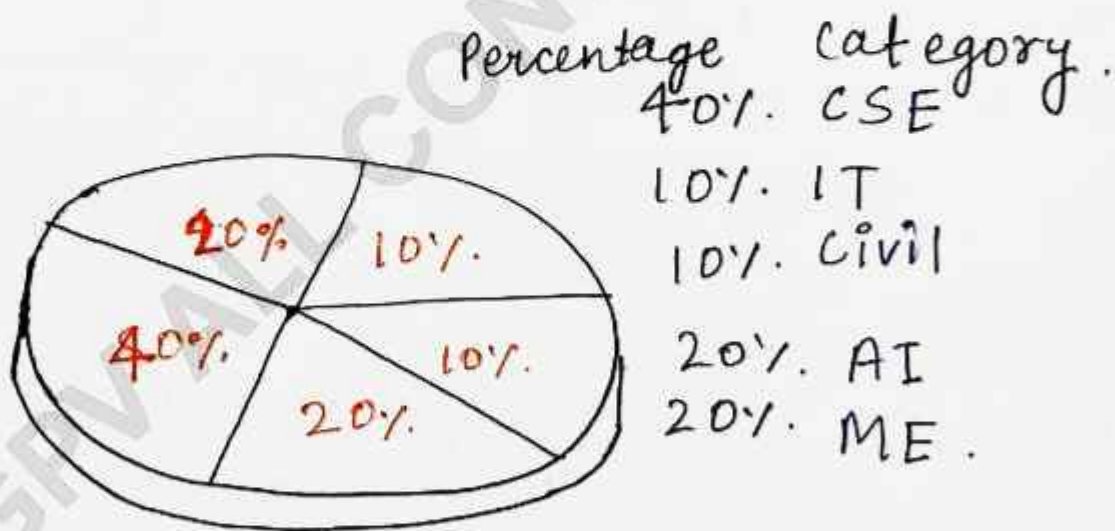


PIE CHART

A Pie chart is a circular graph used to represent data as parts of a whole. The circle is divided into different slices, and each slice shows the percentage or proportion of a category.

A pie chart shows how a whole quantity is divided into different parts.

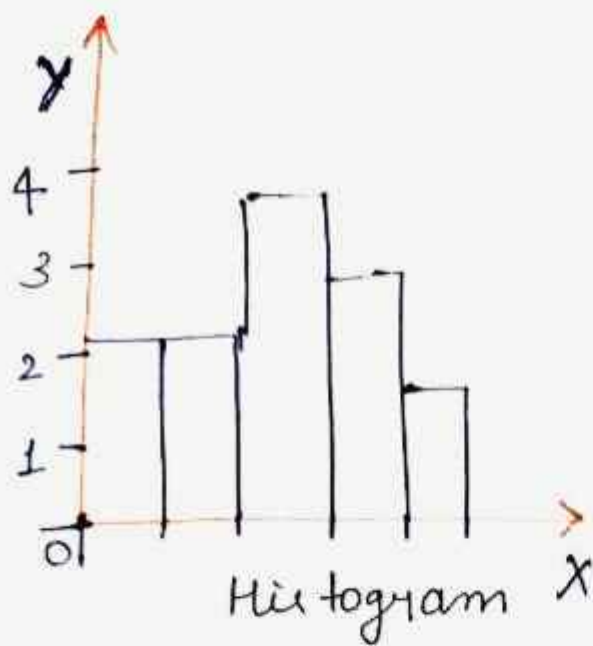
Example. Distribution of students in different streams.



HISTOGRAM

A histogram is a type of graph used to represent the frequency distribution of continuous numerical data using rectangular bars.

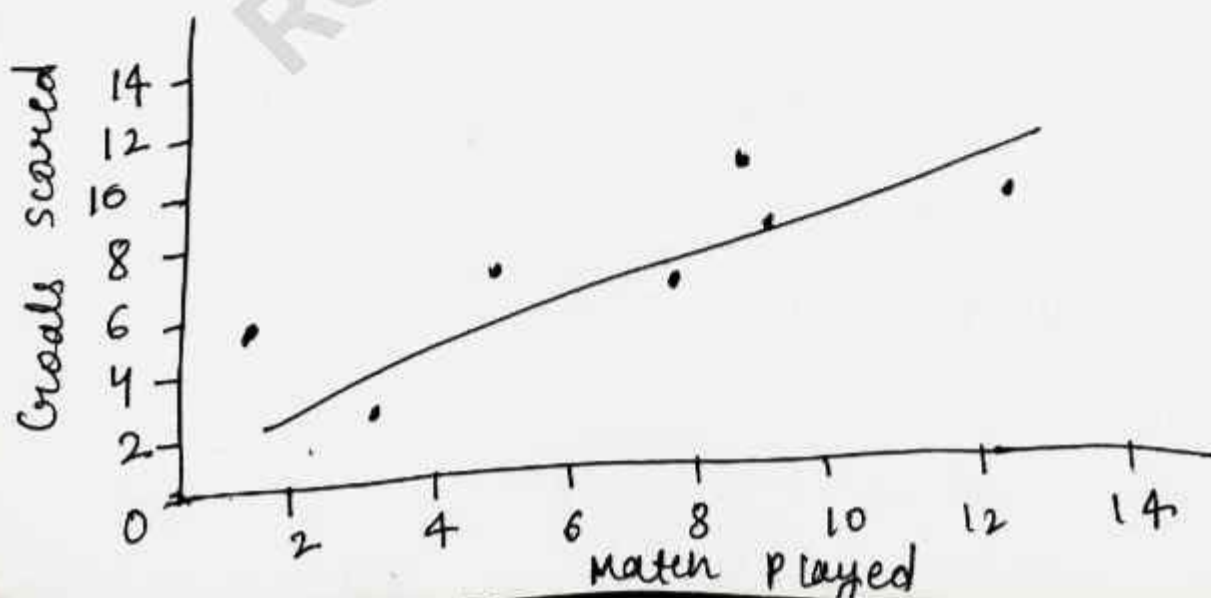
→ A histogram shows how often data values occur within certain ranges (intervals).



SCATTER PLOT.

A scatter plot is a type of graph used to show the relationship between two variables by plotting data points on a coordinate plane.

A scatter plot displays data as points to show how two variables are related.



HYPOTHESIS FUNCTION

A Hypothesis function is a mathematical function used by a machine learning model to predict the output value from input data.

Hypothesis function = a formula used by the model to make predictions.

It represents the relationship between input (features) and output (target values).

Example → Predicting house price based on size:
Input → Size of house
Output → Price of house.

The model uses a hypothesis function to estimate the price.

Example of hypothesis function →
Linear Regression.

HYPOTHESIS TESTING

Hypothesis Testing is a statistical method used to decide whether a statement about data is true or false based on sample data.

Hypothesis testing checks if a claim about data is correct.

Types of Hypothesis.

- ① Null Hypothesis $\rightarrow$ The default assumption that there is no effect or no difference.

Example $\rightarrow H_0$:- The new ML model does not improve accuracy.

- ② Alternative Hypothesis (H_1 or H_a)

The opposite statement that we want to prove.

Example: The new ML model does not improve accuracy.

Example in Machine Learning.

Suppose we build a new ML algorithm and want to test if it performs better.

• $H_0 \rightarrow$ New model accuracy = Old model accuracy.

• $H_1 \rightarrow$ New model accuracy $>$ Old model accuracy.

After testing $\rightarrow$ If result is statistically significant $\rightarrow$ New model is better.

DATA DISTRIBUTION

Data Distribution describes how data values are spread or arranged across different ranges or categories.

Data distribution shows how frequently different values appear in a dataset.

Example $\rightarrow$ Marks of students in a class may be distributed like:-

Marks Range	Number of students
0 - 20	2
21 - 40	5
41 - 60	10
61 - 80	8
81 - 100	3

This table shows how the marks are distributed among students.

Importance of Data Distribution.

Data distribution is important because it helps to :-

- Understand the pattern ~~the~~ of data.
- Identify trends and variations.
- Detect outliers (unusual values)
- Choose the correct machine learning model.
- Perform statistical analysis

DATA PREPROCESSING

Data Preprocessing is the process of cleaning, transforming and organizing raw data before using it in a machine learning algorithms.

It means preparing raw data so that it can be used effectively for machine learning.

Example → If a dataset has missing values, duplicate records, or incorrect formats, preprocessing is used to fix these issues.

Need for Data Preprocessing.

Data Preprocessing is important because real-world data may have.

1. Missing values
2. Noisy data (errors)
3. Duplicate data.
4. Duplicate Data formats.
5. Irrelevant information.

Without preprocessing, machine learning models may produce incorrect predictions.

DATA AUGMENTATION

Data augmentation is a technique used in machine learning and deep learning to increase the size of a dataset by creating new training data from existing data.

It is mainly used when available dataset is small.

It means generating additional data from existing data to improve model training.

Example $\rightarrow$ If we have 100 images of cats, we can create more images by rotating, flipping and cropping.

NORMALIZATION OF DATA SETS.

Normalization is a data preprocessing technique used to rescale numerical features so they fall within a common range.

It converts different - scale data into a uniform scale without changing relationships between values.

The scale of normalization usually between 0 and 1.

Importance of Normalization -

In datasets, features can have very different ranges.

- Age $\rightarrow$ 18 to 60
- Salary $\rightarrow$ 10,000 to 1,00,000.

- Machine Learning models may give more importance to larger values (like salary).
- Normalization fixes this by bringing everything to the same scale.